Course 6044B | Semester VI | Theory | 4 Credits

Modern Electronics

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6044B as Modern Electronics in Semester VI for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on emerging technology curricula and Industry 4.0 frameworks. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. System designs, AI models, network configurations and autonomous algorithms must be based on approved engineering plans, certified hardware data, current industry standards and competent professional review.

Verified source note: Verified sources support Industry 4.0, AI integration, 5G features, and robotics. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Modern Electronics studies the emerging technologies, trends and systems that define the current and future landscape of electronic engineering. It develops the ability to analyze Industry 4.0 concepts, evaluate the integration of Artificial Intelligence (AI) and Robotics, understand 5G and beyond communication, and coordinate innovative electronic solutions while respecting the technical and ethical boundaries of modern engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□.

Prerequisite foundation

Basic electronics, communication systems, microcontrollers and digital signal processing.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the fundamentals of Industry 4.0, smart manufacturing and digitization.
2. Analyze the role of Artificial Intelligence (AI) and Machine Learning in electronic systems.
3. Evaluate the systematic features and applications of 5G and future wireless standards.
4. Explain the emerging trends in robotics, autonomous systems and sustainable electronics.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the principles of Industry 4.0 and smart electronic systems.	Understand
CO2	Analyze the integration of AI and machine learning in modern electronics.	Apply
CO3	Evaluate the features and impact of 5G and future communication trends.	Apply
CO4	Explain emerging trends in robotics, autonomy and green electronics.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Industry 4.0 and Smart Manufacturing	Evolution of Industrial Revolutions; Industry 4.0 pillars: Big Data, Autonomous Robots, Simulation, System Integration, IIoT, Cybersecurity, Cloud, Additive Manufacturing, Augmented Reality.	Approx. 14 hours	CO1	Module 1
2	AI and Machine Learning in Electronics	AI vs Machine Learning vs Deep Learning; Neural network basics; AI on the Edge (TinyML); Applications: Predictive maintenance, Quality control, Voice/Image recognition.	Approx. 16 hours	CO2	Module 2
3	5G and Future Communication Trends	5G NR (New Radio) features: eMBB, URLLC, mMTC; Millimeter wave and Beamforming; Network slicing; Introduction to 6G and Satellite internet (Starlink, Kuiper).	Approx. 16 hours	CO3	Module 3
4	Robotics and Sustainable Electronics	Robotics: Sensors, Actuators, and Control; Collaborative robots (Cobots); Autonomous vehicles basics; Sustainable electronics: E-waste management, Green manufacturing, Energy-efficient design.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Industry 4.0 and Smart Manufacturing

Evolution of Industrial Revolutions; Industry 4.0 pillars: Big Data, Autonomous Robots, Simulation, System Integration, IIoT, Cybersecurity, Cloud, Additive Manufacturing, Augmented Reality.

CO1 · Approx. 14 hours

AI and Machine Learning in Electronics

AI vs Machine Learning vs Deep Learning; Neural network basics; AI on the Edge (TinyML); Applications: Predictive maintenance, Quality control, Voice/Image recognition.

CO2 · Approx. 16 hours

5G and Future Communication Trends

5G NR (New Radio) features: eMBB, URLLC, mMTC; Millimeter wave and Beamforming; Network slicing; Introduction to 6G and Satellite internet (Starlink, Kuiper).

CO3 · Approx. 16 hours

Robotics and Sustainable Electronics

Robotics: Sensors, Actuators, and Control; Collaborative robots (Cobots); Autonomous vehicles basics; Sustainable electronics: E-waste management, Green manufacturing, Energy-efficient design.

CO4 · Approx. 14 hours

MODULE 1

Industry 4.0 and Smart Manufacturing

Evolution of Industrial Revolutions; Industry 4.0 pillars: Big Data, Autonomous Robots, Simulation, System Integration, IIoT, Cybersecurity, Cloud, Additive Manufacturing, Augmented Reality.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

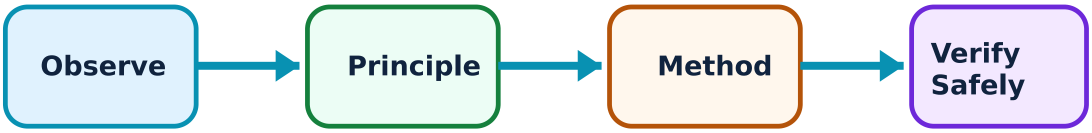
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Industry 4.0 and Smart Manufacturing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Industrial Revolutions and Digitization−

Industry 4.0 represents the fourth industrial revolution, characterized by the 'Cyber-Physical Systems' (CPS). It involves the digitization of manufacturing through horizontal and vertical system integration. Key goals include mass customization, increased efficiency, and real-time decision making.

Study and application method

List the four 'Industrial Revolutions' and their key technologies; define 'Cyber-Physical Systems' (CPS) conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the four 'Industrial Revolutions' and their key technologies; define 'Cyber-Physical Systems' (CPS) conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Rapid digitization increases cybersecurity risks. Do not implement industrial networks without authorised firewalls, encryption, and secure data-handling protocols.

2. Additive Manufacturing (3D Printing)−

Additive manufacturing builds parts layer-by-layer from 3D model data. In electronics, this includes 3D printed circuit boards (3D-PCBs) and custom enclosures. It enables rapid prototyping and complex geometries that are impossible with traditional subtractive methods.

Study and application method

Describe the steps of 'FDM' or 'SLA' 3D printing; list two advantages of '3D Printing' for electronic product development.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the steps of 'FDM' or 'SLA' 3D printing; list two advantages of '3D Printing' for electronic product development.

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Common mistake / precaution: 3D printed parts may have different mechanical properties than molded parts. Do not use 3D printed components for structural or high-stress applications without authorised strength and material testing.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

AI and Machine Learning in Electronics

AI vs Machine Learning vs Deep Learning; Neural network basics; AI on the Edge (TinyML); Applications: Predictive maintenance, Quality control, Voice/Image recognition.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

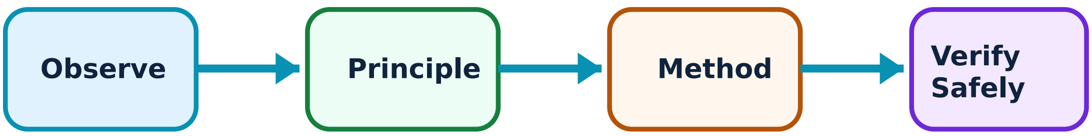
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for AI and Machine Learning in Electronics: identify the system, state its principle, follow the correct method and verify the result safely.

1. AI Fundamentals for Engineers−

Artificial Intelligence (AI) enables machines to mimic human intelligence. Machine Learning (ML) is a subset that uses data to improve performance. Neural networks are inspired by the human brain. In electronics, AI is used for signal processing, pattern recognition, and system optimization.

Study and application method

Differentiate between 'Supervised' and 'Unsupervised' learning conceptually; explain the concept of a 'Neural Network' layer.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Supervised' and 'Unsupervised' learning conceptually; explain the concept of a 'Neural Network' layer.

> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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> **Common mistake / precaution:** AI models can produce biased or incorrect results if trained on poor data. Do not rely solely on AI-based decisions for safety-critical control without authorised human-in-the-loop and fail-safe mechanisms.

2. TinyML and Edge AI–

TinyML involves running machine learning models on resource-constrained microcontrollers. This enables 'Edge AI', where data is processed locally without needing a cloud connection. Applications include keyword spotting, gesture recognition, and anomaly detection in industrial motors.

Study and application method

Define 'TinyML'; list the benefits of 'Edge AI' over cloud-based processing conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'TinyML'; list the benefits of 'Edge AI' over cloud-based processing conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Edge devices have limited memory and power. All AI models for edge deployment must follow authorised quantization and pruning strategies to fit the target hardware.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

5G and Future Communication Trends

5G NR (New Radio) features: eMBB, URLLC, mMTC; Millimeter wave and Beamforming; Network slicing; Introduction to 6G and Satellite internet (Starlink, Kuiper).

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

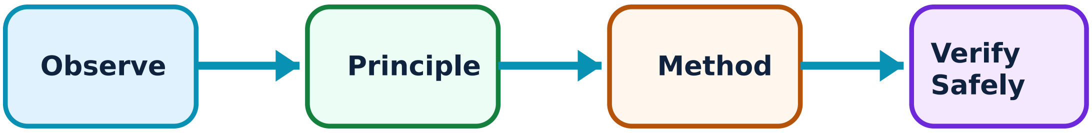
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for 5G and Future Communication Trends: identify the system, state its principle, follow the correct method and verify the result safely.

1. 5G New Radio (NR)−

5G offers three main service categories: Enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and Massive Machine Type Communication (mMTC). It uses higher frequency bands (mmWave) and advanced antenna techniques like Beamforming to increase capacity.

Study and application method

Compare '4G' and '5G' in a conceptual table; explain the purpose of 'Network Slicing' in 5G.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare '4G' and '5G' in a conceptual table; explain the purpose of 'Network Slicing' in 5G.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: 5G mmWave signals have limited range and are easily blocked. All 5G deployments must follow authorised small-cell placement and signal-propagation analysis.

2. Future Connectivity (6G and Satellite)−

6G is expected to integrate AI, Terahertz frequencies, and holographic communication. Satellite internet constellations provide global coverage by using thousands of Low Earth Orbit (LEO) satellites, bridging the digital divide in remote areas.

Study and application method

Describe the concept of 'Satellite Internet' conceptually; list one potential application of '6G' technology.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the concept of 'Satellite Internet' conceptually; list one potential application of '6G' technology.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Satellite constellations increase orbital debris risks. All satellite-based communication must follow authorised orbital regulations and frequency-coordination standards.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Robotics and Sustainable Electronics

Robotics: Sensors, Actuators, and Control; Collaborative robots (Cobots); Autonomous vehicles basics; Sustainable electronics: E-waste management, Green manufacturing, Energy-efficient design.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

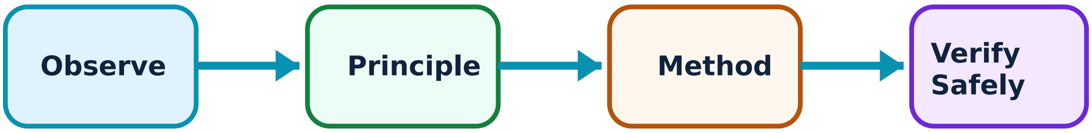
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Robotics and Sustainable Electronics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Robotics and Autonomy−

Modern robotics focuses on 'Cobots' that work alongside humans. Autonomous systems use LiDAR, cameras, and AI to navigate and perform tasks without human intervention. Control algorithms (PID, SLAM) are essential for stability and mapping.

Study and application method

Sketch the components of a 'Robot Control Loop'; explain the role of 'LiDAR' in autonomous navigation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the components of a 'Robot Control Loop'; explain the role of 'LiDAR' in autonomous navigation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Autonomous movements can be unpredictable. Do not operate robots or autonomous systems without authorised safety zones, emergency stops, and collision-avoidance sensors.

2. Green and Sustainable Electronics−

Sustainable electronics focuses on reducing the environmental impact. This includes using recyclable materials, reducing power consumption (Energy Star), and proper E-waste management. Circular economy principles aim to reuse and refurbish electronic components.

Study and application method

Define 'E-waste'; list three 'Green Design' principles for electronic products conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'E-waste'; list three 'Green Design' principles for electronic products conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Hazardous materials in electronics (lead, mercury) require specialized disposal. All E-waste must follow authorised recycling and hazardous-material handling protocols.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Industry 4.0 and Smart Manufacturing

Use the concept flow to revise observation, principle, method and verification.

Module 2: AI and Machine Learning in Electronics

Use the concept flow to revise observation, principle, method and verification.

Module 3: 5G and Future Communication Trends

Use the concept flow to revise observation, principle, method and verification.

Module 4: Robotics and Sustainable Electronics

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Industry 3.0 and Industry 4.0 in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch a 'Neural Network' architecture showing the input, hidden, and output layers.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the '5G Service Triangle' conceptually showing eMBB, URLLC, and mMTC.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the energy saving conceptually for a device moving from cloud-AI to Edge-AI.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a modern electronic technology in your environment (e.g., 5G phone, smart speaker) and note its features.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Industry 4.0 and Smart Manufacturing

Explain Industrial Revolutions and Digitization and state one correct method, application or precaution.−

Industry 4.0 represents the fourth industrial revolution, characterized by the 'Cyber-Physical Systems' (CPS). It involves the digitization of manufacturing through horizontal and vertical system integration. Key goals include mass customization, increased efficiency, and real-time decision making.

Answer framework: List the four 'Industrial Revolutions' and their key technologies; define 'Cyber-Physical Systems' (CPS) conceptually.

Important point: Rapid digitization increases cybersecurity risks. Do not implement industrial networks without authorised firewalls, encryption, and secure data-handling protocols.

Explain Additive Manufacturing (3D Printing) and state one correct method, application or precaution.−

Additive manufacturing builds parts layer-by-layer from 3D model data. In electronics, this includes 3D printed circuit boards (3D-PCBs) and custom enclosures. It enables rapid prototyping and complex geometries that are impossible with traditional subtractive methods.

Answer framework: Describe the steps of 'FDM' or 'SLA' 3D printing; list two advantages of '3D Printing' for electronic product development.

Important point: 3D printed parts may have different mechanical properties than molded parts. Do not use 3D printed components for structural or high-stress applications without authorised strength and material testing.

Module 2 — AI and Machine Learning in Electronics

Explain AI Fundamentals for Engineers and state one correct method, application or precaution.−

Artificial Intelligence (AI) enables machines to mimic human intelligence. Machine Learning (ML) is a subset that uses data to improve performance. Neural networks are inspired by the human brain. In electronics, AI is used for signal processing, pattern recognition, and system optimization.

Answer framework: Differentiate between 'Supervised' and 'Unsupervised' learning conceptually; explain the concept of a 'Neural Network' layer.

Important point: AI models can produce biased or incorrect results if trained on poor data. Do not rely solely on AI-based decisions for safety-critical control without authorised human-in-the-loop and fail-safe mechanisms.

Explain TinyML and Edge AI and state one correct method, application or precaution. −

TinyML involves running machine learning models on resource-constrained microcontrollers. This enables 'Edge AI', where data is processed locally without needing a cloud connection. Applications include keyword spotting, gesture recognition, and anomaly detection in industrial motors.

Answer framework: Define 'TinyML'; list the benefits of 'Edge AI' over cloud-based processing conceptually.

Important point: Edge devices have limited memory and power. All AI models for edge deployment must follow authorised quantization and pruning strategies to fit the target hardware.

Module 3 — 5G and Future Communication Trends

Explain 5G New Radio (NR) and state one correct method, application or precaution.−

5G offers three main service categories: Enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and Massive Machine Type Communication (mMTC). It uses higher frequency bands (mmWave) and advanced antenna techniques like Beamforming to increase capacity.

Answer framework: Compare '4G' and '5G' in a conceptual table; explain the purpose of 'Network Slicing' in 5G.

Important point: 5G mmWave signals have limited range and are easily blocked. All 5G deployments must follow authorised small-cell placement and signal-propagation analysis.

Explain Future Connectivity (6G and Satellite) and state one correct method, application or precaution.−

6G is expected to integrate AI, Terahertz frequencies, and holographic communication. Satellite internet constellations provide global coverage by using thousands of Low Earth Orbit (LEO) satellites, bridging the digital divide in remote areas.

Answer framework: Describe the concept of 'Satellite Internet' conceptually; list one potential application of '6G' technology.

Important point: Satellite constellations increase orbital debris risks. All satellite-based communication must follow authorised orbital regulations and frequency-coordination standards.

Module 4 — Robotics and Sustainable Electronics

Explain Robotics and Autonomy and state one correct method, application or precaution.−

Modern robotics focuses on 'Cobots' that work alongside humans. Autonomous systems use LiDAR, cameras, and AI to navigate and perform tasks without human intervention. Control algorithms (PID, SLAM) are essential for stability and mapping.

Answer framework: Sketch the components of a 'Robot Control Loop'; explain the role of 'LiDAR' in autonomous navigation.

Important point: Autonomous movements can be unpredictable. Do not operate robots or autonomous systems without authorised safety zones, emergency stops, and collision-avoidance sensors.

Explain Green and Sustainable Electronics and state one correct method, application or precaution.−

Sustainable electronics focuses on reducing the environmental impact. This includes using recyclable materials, reducing power consumption (Energy Star), and proper E-waste management. Circular economy principles aim to reuse and refurbish electronic components.

Answer framework: Define 'E-waste'; list three 'Green Design' principles for electronic products conceptually.

Important point: Hazardous materials in electronics (lead, mercury) require specialized disposal. All E-waste must follow authorised recycling and hazardous-material handling protocols.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Industry 4.0 and Smart Manufacturing.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.

3. State one key definition from Module 2: AI and Machine Learning in Electronics.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: 5G and Future Communication Trends.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Robotics and Sustainable Electronics.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Evolution of Industrial Revolutions; Industry 4.0 pillars: Big Data, Autonomous Robots, Simulation, System Integration, IIoT, Cybersecurity, Cloud, Additive Manufacturing, Augmented Reality..
2. Develop a complete answer or practical plan for: AI vs Machine Learning vs Deep Learning; Neural network basics; AI on the Edge (TinyML); Applications: Predictive maintenance, Quality control, Voice/Image recognition..
3. Develop a complete answer or practical plan for: 5G NR (New Radio) features: eMBB, URLLC, mMTC; Millimeter wave and Beamforming; Network slicing; Introduction to 6G and Satellite internet (Starlink, Kuiper)..
4. Develop a complete answer or practical plan for: Robotics: Sensors, Actuators, and Control; Collaborative robots (Cobots); Autonomous vehicles basics; Sustainable electronics: E-waste management, Green manufacturing, Energy-efficient design..

LAST-MILE REVISION

Quick Revision

Module 1: Industry 4.0 and Smart Manufacturing

Evolution of Industrial Revolutions; Industry 4.0 pillars: Big Data, Autonomous Robots, Simulation, System Integration, IIoT, Cybersecurity, Cloud, Additive Manufacturing, Augmented Reality.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: AI and Machine Learning in Electronics

AI vs Machine Learning vs Deep Learning; Neural network basics; AI on the Edge (TinyML); Applications: Predictive maintenance, Quality control, Voice/Image recognition.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: 5G and Future Communication Trends

5G NR (New Radio) features: eMBB, URLLC, mMTC; Millimeter wave and Beamforming; Network slicing; Introduction to 6G and Satellite internet (Starlink, Kuiper).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Robotics and Sustainable Electronics

Robotics: Sensors, Actuators, and Control; Collaborative robots (Cobots); Autonomous vehicles basics; Sustainable electronics: E-waste management, Green manufacturing, Energy-efficient design.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Industrial Revolutions and Digitization	Industry 4.
Additive Manufacturing (3D Printing)	Additive manufacturing builds parts layer-by-layer from 3D model data.
AI Fundamentals for Engineers	Artificial Intelligence (AI) enables machines to mimic human intelligence.
TinyML and Edge AI	TinyML involves running machine learning models on resource-constrained microcontrollers.
5G New Radio (NR)	5G offers three main service categories: Enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and Massive Machine Type Communication (mMTC).
Future Connectivity (6G and Satellite)	6G is expected to integrate AI, Terahertz frequencies, and holographic communication.
Robotics and Autonomy	Modern robotics focuses on 'Cobots' that work alongside humans.
Green and Sustainable Electronics	Sustainable electronics focuses on reducing the environmental impact.

LEARNING RESOURCES

References

Recommended modern electronics references

1. Klaus Schwab, The Fourth Industrial Revolution.
2. Alpaydin Ethem, Introduction to Machine Learning.
3. Afif Osseiran et al., 5G Mobile and Wireless Communications Technology.
4. Saeid Nahavandi, Industry 5.0—A Human-Centric Solution.

Approved public modern electronics sources

- https://nptel.ac.in/courses/108105375
- https://onlinecourses.nptel.ac.in/noc25_ee02/preview

These sources provide the verified study basis while official Course 6044B detail is unavailable; they do not replace official industry designs, authorised technology standards, certified equipment specifications or authorised strategic innovation plans.